



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (CSE) (Credit System) Degree Examinations

Supplementary Examinations - August 2025

21CSE304 - BUSINESS INTELLIGENCE

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*																				
1.	a) Describe the challenges faced to store unstructured data. Explain how UIMA resolves the challenges faced by unstructured data with a neat diagram.	10	L*2	1	1																				
	b) Describe various OLAP architectures.	5	L2	1	1																				
	c) Discuss the following with the help of a case study: i) Roll up ii) Drill down iii) Pivot	5	L2	1	1,2																				
2.	a) In detail discuss the different areas where BI has made its impact.	10	L2	2	1																				
	b) Explain how BI is applied for technology solutions and business solutions	10	L2	2	1																				
3.	a) Discuss various stages of ETL with the help of neat diagram.	10	L2	2	1																				
	b) Explain two approaches of data integration and also discuss common approaches of Data Integration	10	L2	2	1																				
Unit – II																									
4.	a) Explain data profiling in terms of problem and solution.	10	L2	4	1,2																				
	b) Define fact table. Illustrate three types of facts.	10	L2	3	1,2																				
5.	a) List and explain various types of dimension tables.	10	L2	3	1																				
	b) Briefly discuss dimensional modeling life cycle with a neat diagram.	10	L2	3	1																				
6.	a) Illustrate any 10 salient attributes of a good metric.	10	L2	4	2																				
	b) Why enterprises need dashboards? Describe the steps for creating dashboards.	10	L2	4	1,2																				
Unit – III																									
7.	a) What is data mining? Explain the process of extracting knowledge from data.	10	L2	5	1																				
	b) Explain Data characterization and Data discrimination. Discuss how other kinds of data can be mined.	10	L2	5	1																				
8.	a) Generate candidate itemsets and frequent itemsets for below table, where the minimum support count is 2.																								
	<table><tr><th>TID</th><th>List of item IDs</th></tr><tr><td>1</td><td>I1, I2, I5</td></tr><tr><td>2</td><td>I2, I4</td></tr><tr><td>3</td><td>I2, I3</td></tr><tr><td>4</td><td>I1, I2, I4</td></tr><tr><td>5</td><td>I1, I3</td></tr><tr><td>6</td><td>I2, I3</td></tr><tr><td>7</td><td>I1, I3</td></tr><tr><td>8</td><td>I1, I2, I3, I5</td></tr><tr><td>9</td><td>I1, I2, I3</td></tr></table>	TID	List of item IDs	1	I1, I2, I5	2	I2, I4	3	I2, I3	4	I1, I2, I4	5	I1, I3	6	I2, I3	7	I1, I3	8	I1, I2, I3, I5	9	I1, I2, I3				
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	b) What is Apriori algorithm? Show the steps of the Apriori algorithm for discovering frequent item sets in Boolean association rule mining.	10	L2	5	2																				
		10	L2	5	2																				

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
